

Big Data (Friday)

HIVE: Apache Hive

It is open source data warehouse system built on top of Hadoop.

It is used for querying and analyzing large datasets stored in Hadoop files.

It processes structured and semi-structured data in Hadoop.

It uses language called HiveQL (HQL) which is similar to SQL. HiveQL automatically translates SQL-like queries into

MapReduce jobs.

It abstracts the complexity of Hadoop.

It organizes data into tables.

Facts of Hive

Hive is not a relational database.

Hive is not a design for Online Transaction Processing (OLTP).

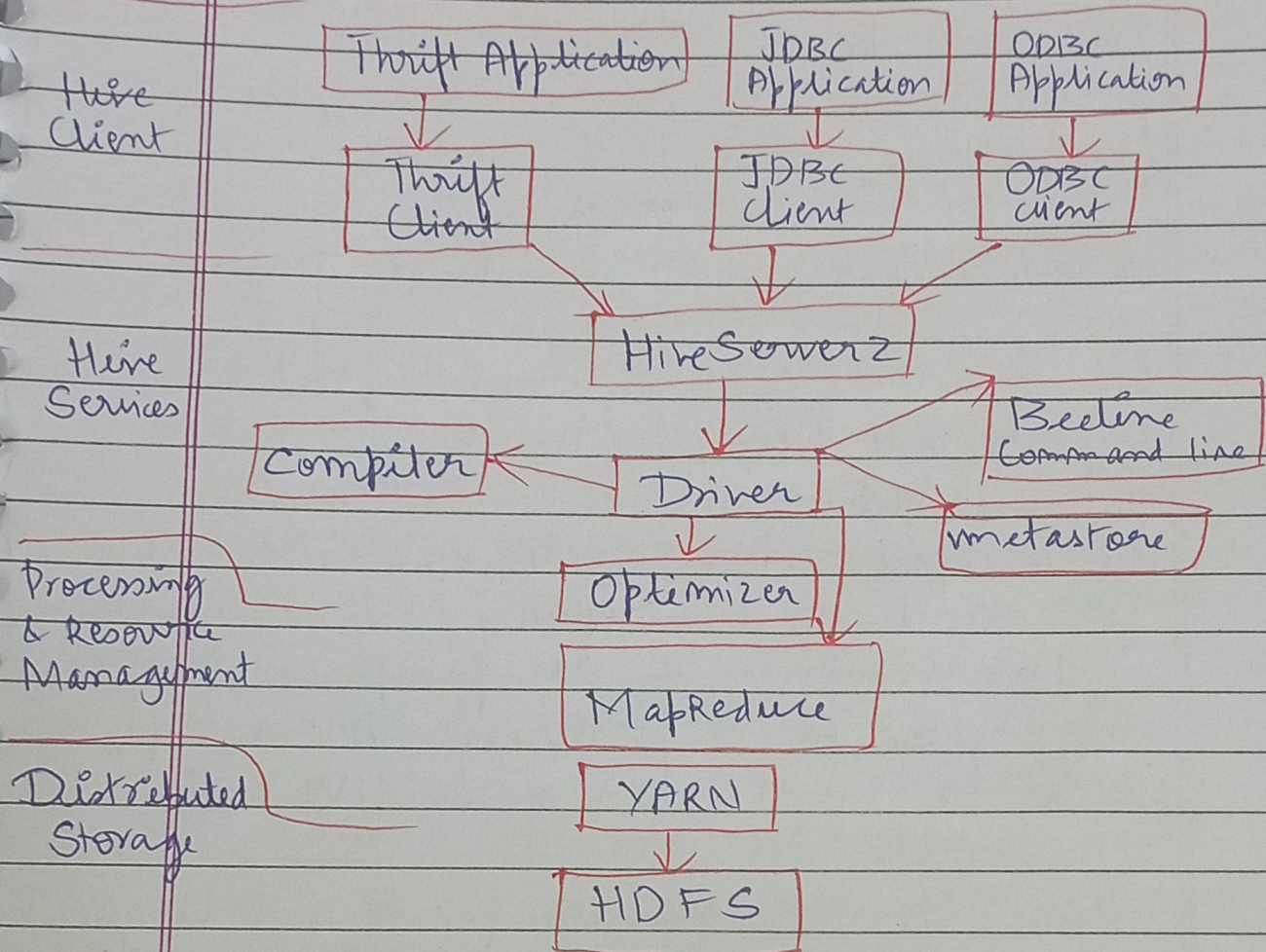
Hive is not a language for real-time queries and row-level updates.

We use Hive for Schema flexibility as well as evolution.

We can use JDBC/ODBC drivers, since they are available in Hive.

(J - Java
O - Object)

Hive Architecture



Hive Clients

Apache hive supports all applications written in language like C++, Java, Python, etc, using JDBC, Thrift and ODBC drivers. Thus, one can easily write hive client application written in a language of their choice. The hive supports different types of Client applications for performing queries. These clients are categorized into 3 types.

- ① Thrift Clients - As Apache hive server is based on thrift, so it serve the Req. from all those languages that support thrift.

② JDBC Clients - Apache Hive allows Java applications to connect to it using JDBC driver

③ ODBC Clients - ODBC driver allows application that support ODBC protocol to connect to Hive. For eg JDBC driver, ODBC uses Thrift to communicate with the Hive server.

CLI or Beeline (command line Interface) -

Default shell Hive provides, u can execute ur hive queries and command directly.

Web Interface - Hive provides web based GUI for executing hive queries and commands

Hive Server - built of Apache Thrift also called as Thrift Server. Allows diff. Client to submit Req's to Hive and retrieve the final Result.

Hive Driver - Responsible for receiving the queries submitted Thrift, JDBC, ODBC, CLI, Web UI interface by Hive Client. This starts the execution of statement, creating sessions. Monitors the life cycle and progress of execution. This stores necessary Metadata generate during execution of Hive SQL statement. Acts as a collection point of data or query result after Reduce operation.

Compiler - Hive driver passes query to compiler. Parsing, type checking, semantic analysis takes place with help of schema present in metastore.

Optimizer - generates optimized logical plan in form of DAG (Directed Acyclic Graph), of MapReduce and HDFS tasks. Aggregate transformations together, as converting pipeline of joins to a single join, for better performance.

Optimizer can also split the tasks, such as applying a transformation on data before reduce operation, to provide better performance.

Executor - compilation and optimization complete, execution engine execute these tasks in order of their dependencies using Hadoop.

Meta Store - Central Repository of Apache Hive metadata in Hive Architecture.

Stores Metadata for Hive tables (like schema, col. in a table, data types and location) and partitions in relational DB.

Provides client access to this info by using metastore service API. Hive metastore consists of 2 fundamental units:

- Service provides metastore access to other Apache Hive Services.
- Disk storage for Hive metadata which is separate from HDFS storage.

Hive Partition Table

BDA (6/01/20)

Load Data in Hive Partitions

```
CREATE TABLE Employee  
(ID BIGINT, NAME STRING, AGE INT, SALARY BIGINT)  
PARTITIONED BY (DEPARTMENT STRING)  
ROW FORMAT DELIMITED FIELDS TERMINATED BY ','  
LINES TERMINATED BY '\n'  
STORED AS TEXTFILE;
```

Employee table stored as TEXTFILE PARTITIONED BY
DEPARTMENT

Input Files:-

Suppose we have 2 DEPARTMENTS - HR and finance

Contents of '/home/hadoop/hr-employees.csv'

1, aarti, 28, 30000

2, sakshi, 22, 22000

3, pra, 50, 15000

Contents of '/home/hadoop/finance-employees.csv'

- Load employees data for partition having department
as HR

```
LOAD DATA INPATH '/home/hadoop/hr-employees.csv'  
INTO TABLE Employee PARTITION (department = 'HR');
```

- Load employees data for partition having department
as Finance

```
LOAD DATA INPATH '/home/hadoop/finance-employees.csv'  
INTO TABLE Employee PARTITION (department =  
'finance');
```

once data is loaded in the table partitions, we can
see Hive has created two directories under
the Employee table directory on.

Query on Partition having value as 'HR'

select * from employee where department = 'HR';

Query on Partition having value as 'Finance'

select * from employee where department = 'Finance';

id	name	age	salary	department
10001	arshi	28	30000	HR
10002	sakshi	22	20000	HR
10003	rajesh	20	10000	Finance

Query on the complete data

select * from employee

We see query for particular partition, reads data from partition only and therefore the queries on a set of partitions perform fast on partitioned tables. In the query result, the partition column and its value is added as last column. Choosing right columns to partition the table is the major + csk as this will greatly impact the query performance.

Insert a row into an employee table

Insert into employee partition (department = 'HR') values (4, 'rani', 40, 20000);

Use INSERT OVERWRITE TABLE to delete existing data in a partition and load with new data.

Insert overwrite table employee partition (department = 'HR') values (4, 'raj', 40, 20000);

Hive Bucketing table

Create a employee table stored as text file partitioned by DEPARTMENT and clustered by salary

```
CREATE TABLE Employee
```

```
(ID BIGINT, NAME STRING, AGE INT,  
SALARY BIGINT)
```

```
PARTITIONED BY (DEPARTMENT STRING)
```

```
CLUSTERED BY SALARY INTO 4 BUCKETS
```

```
ROW FORMAT DELIMITED FIELDS TERMINATED  
BY ','
```

```
LINES TERMINATED BY '\n'
```

```
STORED AS TEXTFILE;
```

Basic Operation on Hive Tables

To show partition:

```
Show partitions Employee;
```

To show tables:

```
Show Tables;
```

To Describe Table

```
Describe Employee;
```

To Identify the Partition Key

```
Describe Extended Employee;
```

To find the location of internal and external tables

Describe Formatted Employee;

Test 12/3/26

It highlighted the column names where
it shows all partition column of the table
and locations where partitions will
store.

Sec 2 Complete
Until Hive

Hive - Built-in Operators

- Relational operators
- Arithmetic operators
- Logical operators
- Complex operators

Relational operators

- $A = B$
- $A \neq B$
- $A < B$
- $A \leq B$
- $A > B$
- $A \geq B$
- $A \text{ IS NULL}$
- $A \text{ IS NOT NULL}$
- $A \text{ LIKE } B$ (Comparison)
- $A \text{ RLIKE } B$
- $A \text{ REGEXP } B$

Arithmetic operators

- $A + B$
- $A - B$
- $A * B$
- A / B
- $A \% B$
- $A \& B$
- $A | B$
- $A \wedge B$
- $\sim A$

Logical operators

$A \text{ AND } B$

$A \&\& B$

$A \text{ OR } B$

$A || B$

$\text{NOT } A$

$!A$

Complex Operators

- $A[n]$
- $M[\text{key}]$
- $S \cdot X$

Hive - Built-in Functions

Function	Signature	Return Type
round()	round(double a)	BIGINT
	SELECT round(21.6) from temp;	
floor()	floor(double a)	BIGINT
	SELECT floor(21.6) from temp;	
ceil()	ceil(double a)	BIGINT
	SELECT ceil(21.6) from temp;	
rand()	rand()	double
	SELECT rand() from temp;	
concat()	concat(string A, string B, ...)	string
	SELECT concat("Hello", "Students") from temp;	
substr()	substr(string A, int start)	string
	SELECT substr("Hello students", 6) from temp;	
substr()	substr(string A, int start, int length)	string
	SELECT substr("Hello Students", 6, 10) from temp;	
upper(), ucase()	upper(string A), ucase (string A)	string
lower(), lcase()	lower(string A), lcase(string A)	string

trim()	trim(string A)	string
ltrim()	ltrim(string A)	string
rtrim()	rtrim(string A)	string

trim - remove whitespaces, ltrim - remove whitespaces from left
 & rtrim - remove from right

to date()	to-date(string timestamp)	string
year()	year(string date)	int
month()	month(string date)	int
day()	day(string date)	int

SELECT to-date("30-09-2023") from temp;

size()	size(Map <K,V>)	int
size()	size(Array <T>)	int
cast()	cast(<expr> as <type>)	value of <type>

SELECT cast('1' as BIGINT) from temp;

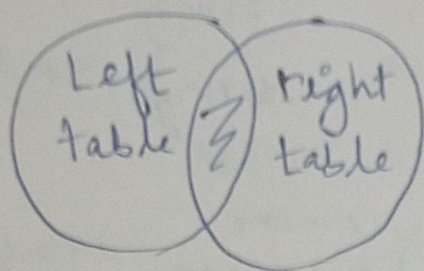
high Aggregate functions

count()	count(*), count(expr)	BIGINT
sum()	sum(col), sum(DISTINCT col)	DOUBLE
avg()	avg(col), avg(DISTINCT col)	DOUBLE
min()	min(col)	DOUBLE
max()	max(col)	DOUBLE

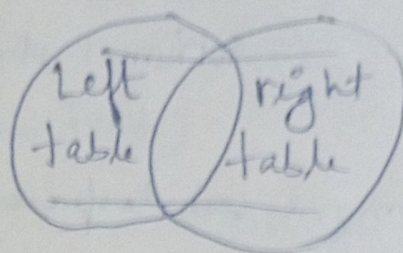
Apache Hive Join

For combining specific fields from two tables by using values common to each one we use Hive Join

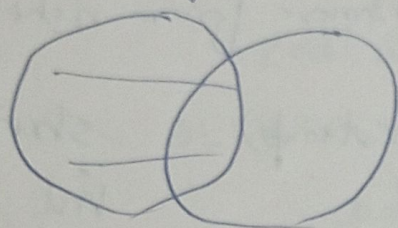
INNER JOIN



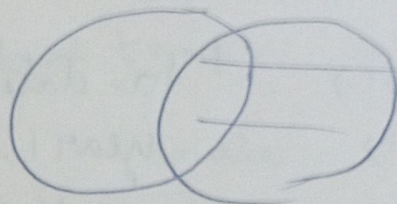
FULL JOIN



Left Join



Right Join



Inner Join

To Combine and retrieve the records from Multiple tables we use HIVE Join Clause

```
hive> SELECT C.ID, C.NAME, C.AGE, O.AMOUNT  
FROM CUSTOMERS C JOIN ORDERS O ON (C.ID = O.CUSTOMER-ID);
```

Left Outer Join

it returns all values from left table, Also Matched values from right table or NULL in case of no matching JOIN Predicate.

```
hive > SELECT C.ID, C.NAME, O.AMOUNT, O.DATE  
FROM CUSTOMERS C LEFT OUTER JOIN ORDERS O ON  
(C.ID = O.CUSTOMER-ID);
```

Right Outer Join

it returns all the values from the right table. Also, the Matched values from the left table or NULL in case of no matching join predicate

hive > SELECT c.ID, c.NAME, o.AMOUNT, o.DATE FROM

Full Outer Join

It combines the records of both the left and the right outer tables which fulfill the join condition. Moreover this joined table contains either all the records from both the tables or fills in NULL values for missing matches on either side.

Hive-QL - Select - Where

SELECT [ALL | DISTINCT] select_expr, select_expr, ...

FROM table-reference

[WHERE where-condition]

[GROUP BY col-list]

[HAVING having-condition]

[SORT BY col-list]]

[LIMIT number] ;

hive > SELECT * FROM employee WHERE salary > 40000;

hive > SELECT * FROM employee ORDER BY DEPT;

hive > SELECT DEPT, count(*) FROM employee
GROUP BY DEPT;

hive > SELECT DEPT, sum(salary) FROM employee
GROUP BY DEPT;

Making Inference

Inference is the task of Making statements about things u can't observe

It refers to the process of taking info u already know

To make inference from data - u need to follow 3 or 4 simple steps

1. Identify the population for which u want to make inferences
2. As u have identified the population, its time to talk about what ur making inference about
3. Now u need to describe sampling process of how u end up with that data as it will tell u have understood the inferences or not.
If u cant explain process it will be very hard for u to understand how valid ur inferences are from data
4. To make inference better u can characterize ur population

What is the diff. Bln drawing inference and making predictions from the data?

Inference from data is generally a backward analysis of past. Prediction from data is a forward analysis for future. Inference is generally deterministic and can be verified to be true / false. Prediction is probabilistic and can't be verified to be 100% true or false

Step by Step Process to Make Inference from data.

First step - u need to do after collecting is to review and process collected data so u can apply statistical analysis to it. The statistical analysis process helps researcher

to extd seq. info from initial data and to generalize results if necessary.

- ① Organize the dataset in Ascending order
- ② Then find the Least and greatest values, Median, lower and upper quartiles
- ③ Calculate the upper (Q_3) and lower Quartile (Q_1) by medians of each "half" of the no. line that includes all the data
- ④ Then u should draw a no. line that includes all the data values
- ⑤ In the next step, u should plot a point for each of the values found (least and greatest values, median, lower and upper quartiles)
- ⑥ From lower to upper quartile draw a box. Inside box draw a vertical line through median
- ⑦ In last step, u should draw whisker by connecting least and greatest values to the box

Ex1 problem

The no. of pets owned by a random sample of students at park Middle school. Use data to make box plot. How Many pets does every student have

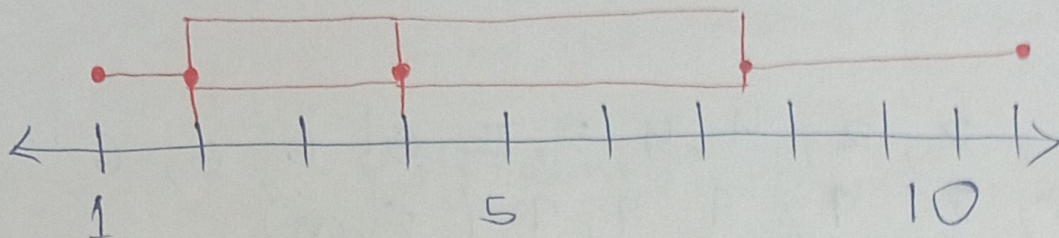
2, 4, 1, 5, 9, 4, 2, 11, 6

Order from least to greatest 1, 2, 2, 4, 4, 5, 6, 9, 11
Least value = 1
Greatest value = 11

(Q2) Median $\rightarrow$ middle value of dataset $1, 2, 2, 4, 4, 5, 6, 9, 11 = 4$

Q1 $\rightarrow$ Median of lower half $1, 2, 2, 4 = (2+2)/2 = 2$

Q3 = $5, 6, 9, 11 = (6+9)/2 = 7.5$
Median of upper half



Almost every student has 1 pet

② Jack collected random sample of 130 Klomen regarding Klomen's color bag preference for his new bag business. Make an inference from the data

Sample	Black Bag	Green B	Blue B	Total
# 1	29	27	74	130

Solⁿ. Inference: According to table above, most people prefer Blue Bags

③ Management wants to open cafeteria in office Building so they collected data from random sample of 50 employees regarding employee's office lunch preference. Make at least 2 inferences based on Result

Cheesesteak	Corn dog	Sandwich	Total
25	12	63	100

Most employee prefer Sandwich
Corn dog least popular

④ Draw a box plot for the data set

{ 1, 6, 9, 2, 6, 6, 5, 8, 1, 9, 4, 1, 6, 9, 2, 6, 6, 5, 8, 1, 9, 4 }

= 22

1, 1, 1, 1, 2, 2, 4, 4, 5, 5, 6, 6, 6, 6, 6, 6, 8, 8, 9, 9, 9, 9 (Even)

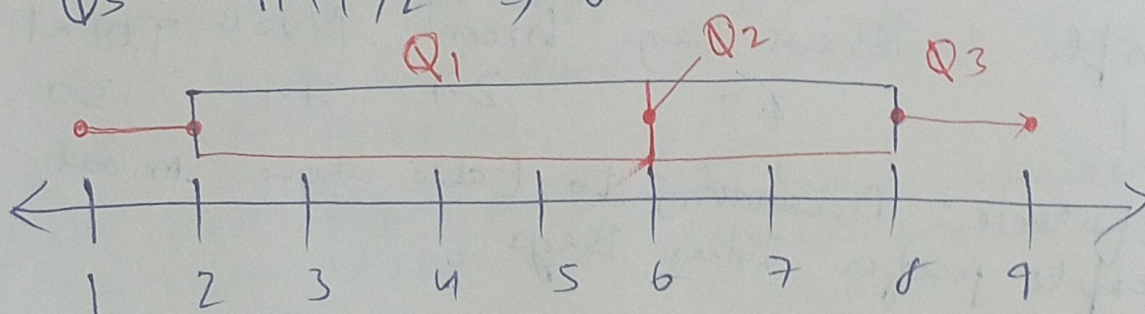
Smallest = 1

Largest = 9

$$Q_2 = \underline{6} = n/2 \Rightarrow 22/2 \Rightarrow 11^{\text{th}} \Rightarrow 6$$

$$Q_1 = \underline{2} \text{ 6th element } \Rightarrow 11+1/2 \Rightarrow 2$$

$$Q_3 = 11+1/2 \Rightarrow 6^{\text{th}} = 8$$



⑤ 14, 5, 15, 7, 8, 11, 13, 14, 6, 14, 5, 15, 7, 8, 11, 13, 14, 6

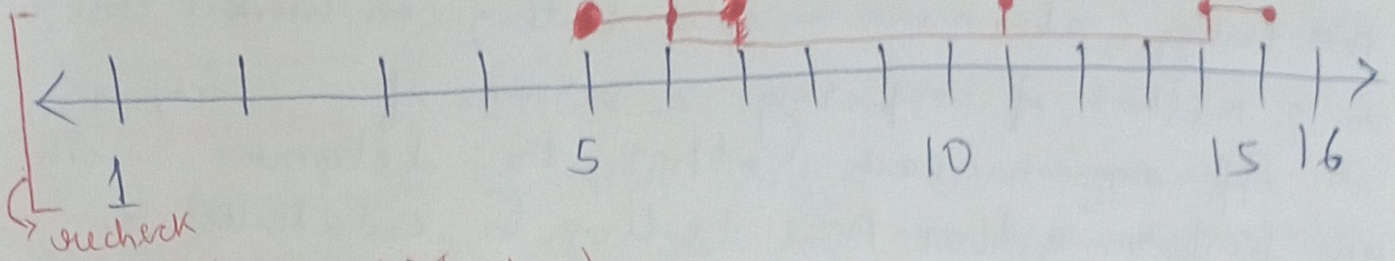
5, 5, 6, 6, 7, 7, 8, 8, 11, 11, 13, 13, 14, 14, 14, 14, 15, 15 $\Rightarrow n = 18$

$$\frac{18}{2} = 9$$

$$Q_2 \Rightarrow 11$$

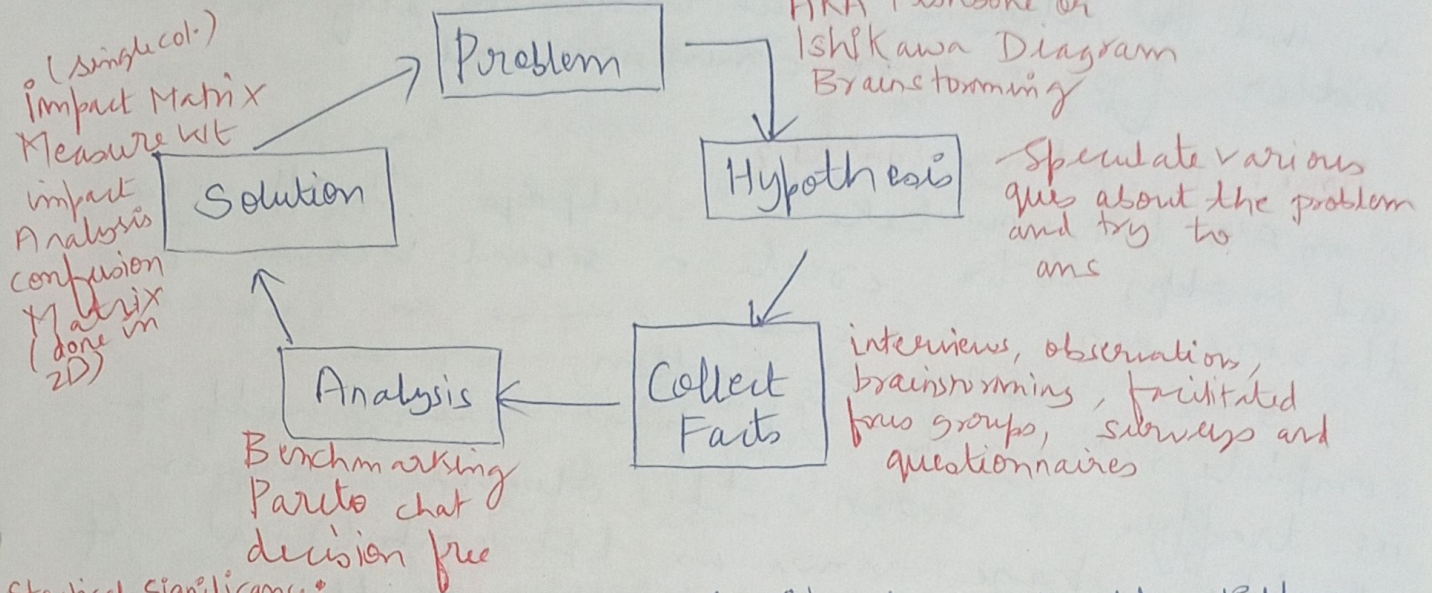
$$Q_1 \Rightarrow 8 \text{ 9th element } \Rightarrow 8$$

$$Q_3 \Rightarrow 9+1/2 = 14$$



B.D.A 9th March (Monday)

Analytical Approach to find solutions for problems: A point Solution



Statistical Significance:

Statistical significance is the likelihood that the difference in conversion rates b/n a given variation and the baseline is not due to random chance. Or statistical significance level reflects or risk tolerance and confidence level.

For ex if u run an A/B testing experiment with a significance level of 95% this means that if u determine a winner, u can be 95% confident that the observed results are real and not an error caused by randomness. It also means that there is a 5% chance that u could be wrong.

Online web owners, marketers, and advertisers have recently become interested in making sure their a/b test experiments get statistical significance before jumping to conclusions.

What is A/B testing?

How A/B Testing Works ?

In an A/B test, you take a webpage or app screen and modify it to create a second version of the same page. This change can be as simple as a single headline or button, or be a complete redesign of the page. Then, half of your traffic is shown the original version of the page (known as the control) and half are shown the modified version of the page (the variation).

Statistical Hypothesis Testing

- **P-value** refers to the probability value of observing an effect from a sample. A p-value of < 0.05 is the conventional threshold for declaring statistical significance.
- **Confidence interval around effect size** refers to the upper and lower bounds of what can happen with your experiment.

The conversion rate is the number of conversions divided by the total number of visitors. For example, if an ecommerce site receives 200 visitors in a month and has 50 sales, the conversion rate would be 50 divided by 200, or 25%.

- A conversion can refer to any desired action that you want the user to take. This can include anything from a click on a button to making a purchase and becoming a customer. Websites and apps often have multiple conversion goals, and each will have its own conversion rate.

Why Is Statistical Significance Important for Business?

Statistical significance is important because it gives you confidence that the changes you make to your website or app actually have a positive impact on your conversion rate and other metrics. Your metrics and numbers can fluctuate wildly from day to day, and statistical analysis provides a sound mathematical foundation for making business decisions and eliminating false positives.

Statistical Hypothesis Testing

- For example, if 100 people visit a website and all the men are shown one version of a page and all the women are shown a different version, then a comparison between the two is not possible, even if the traffic is split 50-50, because the difference in demographics could introduce variations in the data. A truly random sample is needed to determine that the result of the experiment is statistically significant.
- In the pharmaceutical industry, researchers use statistical test results from clinical trials to evaluate new drugs.

Statistical Hypothesis Testing

- Statistical significance is most practically used in statistical hypothesis testing. For example, you want to know whether or not changing the color of a button on your website from red to green will result in more people clicking on it.
- If your button is currently red, that's called your "null hypothesis". Turning your button green is known as your "alternative hypothesis". To determine the observed difference in a statistical significance test, you will want to pay attention to two outputs: p-value and confidence interval around effect size.

Key Technologies that enable Big Data Analytics for Business



Predictive Analytics



NoSQL Databases



Knowledge Discovery Tools



Stream Analytics



In-memory Data Fabric



Distributed Storage



Data Virtualization



Data Integration



Data Preprocessing



Data Quality